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Practical 10 Final Task

Code :

```
<?pr10.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <title>Student Result Analytics</title>
6    <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
7    <link rel="stylesheet" href="style.css">
8    <style>
9      body {
10        font-family: Arial, sans-serif;
11        background-color: #f5f6fa;
12        text-align: center;
13        padding: 20px;
14      }
15      table {
16        margin: 20px auto;
17        border-collapse: collapse;
18        width: 80%;
19        background-color: #fff;
20        box-shadow: 0 0 10px rgba(0,0,0,0.1);
21      }
22      th, td {
23        border: 1px solid #ddd;
24        padding: 10px;
25      }
26      th {
27        background-color: #3498db;
28        color: white;
29      }
30      tr:nth-child(even) {
31        background-color: #f9f9f9;
32      }
33      tr.topper {
34        background-color: #c6f7c3 !important; /* highlight toppers */
35        font-weight: bold;
36      }
37      button {
38        padding: 10px 15px;
39        margin: 10px;
40        background-color: #3498db;
41        color: white;
42        border: none;
43        border-radius: 5px;
44        cursor: pointer;
45      }
46      button:hover {
47        background-color: #2980b9;
48      }
49      #avgContainer {
50        margin-top: 20px;
51        font-weight: bold;
52        color: #2c3e50;
53      }
54    </style>
55  </head>
56  <body>
57
58    <h1>📊 Student Result Analytics Dashboard</h1>
59    <button id="loadFetch">Load Data (Fetch API)</button>
60    <button id="loadjQuery">Load Data (jQuery AJAX)</button>
61    <button id="showAvg">Show Subject Averages</button>
62
63    <table id="studentTable">
64      <thead>
65        <tr>
66          <th>ID</th>
67          <th>Name</th>
68          <th>Subject</th>
69          <th>Marks</th>
70          <th>Grade</th>
71        </tr>
72      </thead>
73      <tbody></tbody>
74    </table>
75
76    <div id="avgContainer"></div>
77
78    <script src="script.js"></script>
79  </body>
80 </html>
81
```

```

JS script.js > ...
1   let studentData = [];
2
3   // Function to render data in the table
4   function displayData(data) {
5       const tbody = document.querySelector("#studentTable tbody");
6       tbody.innerHTML = "";
7
8       data.forEach(stu => {
9           const row = document.createElement("tr");
10
11           // Highlight toppers (marks > 90)
12           if (stu.marks > 90) {
13               row.classList.add("topper");
14           }
15
16           row.innerHTML = `
17               <td>${stu.id}</td>
18               <td>${stu.name}</td>
19               <td>${stu.subject}</td>
20               <td>${stu.marks}</td>
21               <td>${stu.grade}</td>
22           `;
23           tbody.appendChild(row);
24       });
25   }
26
27   // FETCH API implementation
28   document.getElementById("loadFetch").addEventListener("click", () => {
29       fetch("students.json")
30           .then(response => response.json())
31           .then(data => {
32               studentData = data;
33               displayData(data);
34               document.getElementById("avgContainer").innerHTML = "";
35           })
36           .catch(err => console.error("Fetch Error:", err));
37   });
38
39   // jQuery AJAX implementation
40   $("#loadjQuery").click(() => {
41       $.getJSON("students.json", function(data) {
42           studentData = data;
43           displayData(data);
44           $("#avgContainer").html("");
45       }).fail(() => alert("Error loading data using jQuery!"));
46   });
47
48   // Calculate subject averages using reduce()
49   document.getElementById("showAvg").addEventListener("click", () => {
50       if (studentData.length === 0) {
51           alert("Please load data first!");
52           return;
53       }
54
55       // Group by subject
56       const subjectGroups = {};
57       studentData.forEach(stu => {
58           if (!subjectGroups[stu.subject]) {
59               subjectGroups[stu.subject] = [];
60           }
61           subjectGroups[stu.subject].push(stu.marks);
62       });
63
64       // Calculate averages using reduce()
65       const averages = {};
66       for (let subject in subjectGroups) {
67           const marksArr = subjectGroups[subject];
68           const avg = marksArr.reduce((sum, m) => sum + m, 0) / marksArr.length;
69           averages[subject] = avg.toFixed(2);
70       }
71
72       // Display averages
73       let avgHTML = "<h3> Subject Averages:</h3>";
74       for (let sub in averages) {
75           avgHTML += `<p>${sub}: ${averages[sub]}</p>`;
76       }
77       document.getElementById("avgContainer").innerHTML = avgHTML;
78   });
79

```

```
{ } students.json > { } 4 >  name
1  [
2    { "id": 1, "name": "Sanika", "subject": "Maths", "marks": 88, "grade": "A" },
3    { "id": 2, "name": "Adhi", "subject": "Science", "marks": 76, "grade": "B" },
4    { "id": 3, "name": "Sara", "subject": "Maths", "marks": 95, "grade": "A+" },
5    { "id": 4, "name": "Shreyali", "subject": "English", "marks": 67, "grade": "C" },
6    { "id": 5, "name": "MAdhura", "subject": "Science", "marks": 92, "grade": "A+" }
7  ]
8
```

Output:



Student Result Analytics Dashboard

Load Data (Fetch API)

Load Data (jQuery AJAX)

Show Subject Averages

ID	Name	Subject	Marks	Grade
1	Sanika	Maths	88	A
2	Adhi	Science	76	B
3	Sara	Maths	95	A+
4	Shreyali	English	67	C
5	MAdhura	Science	92	A+



Student Result Analytics Dashboard

Load Data (Fetch API)

Load Data (jQuery AJAX)

Show Subject Averages

ID	Name	Subject	Marks	Grade
1	Sanika	Maths	88	A
2	Adhi	Science	76	B
3	Sara	Maths	95	A+
4	Shreyali	English	67	C
5	MAdhura	Science	92	A+



Subject Averages:

Maths: 91.50

Science: 84.00

English: 67.00